Feng CHEN

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SUMMARY

PhD candidate in Health Informatics at the University of Washington working at the intersection of natural language processing, large language models, and human-centered AI. Research spans LLM and agent evaluation, robustness and bias auditing, and speech- and language-based assessment of mental health, with 12 peer-reviewed papers in venues including JAMIA, Neuropsychologia, Cell Patterns, ACL/NeurIPS workshops, CHIL, AMIA, and PSB. Industry experience as an Applied Scientist Intern at Microsoft and a Software Development Engineer Intern at Amazon, with end-to-end ownership of data pipelines, model development, large-scale evaluation, and statistical analysis.

EDUCATION

University of Washington Seattle, WA Sep 2023 – now
PhD candidate in Health Informatics **Cumulative GPA:** 3.90

Harvard Medical School Longwood, MA Sep 2021 – Dec 2022
Master of Biomedical Informatics **Cumulative GPA:** 3.96

Brandeis University Waltham, MA Jan 2018 – Dec 2020
Double major: Biochemistry & Computer Science **Cumulative GPA:** 3.59 (Dean's list) **GPA (Math & CS):** 3.78
Award: Jacques Cohen Award in Computer Science (awarded for excellent researchers)
Main Coursework: Deep Learning for Natural Language Processing, Data Structures and Algorithms, Statistical Machine Learning, Deep Learning for Biomedical Data, Clinical Informatics, Data Visualization, Precision Medicine, Data Science for Medical Decision Making

SKILLS

Programming: Python (proficient), Java (proficient), SQL, R, Scheme, MATLAB, HTML
ML & NLP stack: PyTorch, Hugging Face Transformers, scikit-learn, Whisper (ASR), Pyannote (diarization), pandas/NumPy
Software and tools: AWS, Git/GitHub, Linux, CLI, Claude Code, Microsoft Office, Cortellis, Photoshop
LLM & evaluation: prompt engineering, multi-agent LLM pipelines, benchmark and eval-harness design, controlled variant generation, robustness and fairness auditing, statistical validation
Research skills: large language models; machine learning; deep learning; natural language processing; clinical data processing; automatic speech recognition; bioinformatics analysis; data cleaning, analysis and visualization; PRISMA systematic review; qualitative coding; proposal and manuscript writing; paper reviewing

RESEARCH EXPERIENCES

University of Washington | Advisor: Prof. Trevor Cohen Seattle, WA
Research assistant

NLP for Social Signals Detection in Patient-Provider Clinical Dialogs Sep 2023 – now

- Utilized ASR (Whisper) and speaker diarization (Pyannote) to transcribe and analyze patient-clinician conversations.
- Trained and compared NLP and LLM models to classify social signal scores for patients and clinicians.
- Investigated implicit racial bias in clinical dialogs and optimized predictive performance for detecting social signals.

Data-driven Clinical Signatures for People with Hallucinations Mar 2024 – now

- Building an automated pipeline to derive multimodal data-driven clinical signatures from mobile behavioral tasks to predict individual differences in severe negative outcomes among people experiencing hallucinations.
- Developed and improved an automated sentence-coherence measurement package to evaluate audio recordings of verbal recall tasks; combined pause dynamics and semantic coherence for automated assessment of thought disorder (published in Neuropsychologia).

- Identifying and mitigating bias in automatic speech recognition and predictive models across demographic groups.

Comparative Analysis of LLMs for Mental Illness Detection in Interviews Jan 2025 – now

- Developed and compared transformer-based and embedding-based NLP models (e.g., Mental-RoBERTa, SentenceBERT, LLaMA) for PTSD and depression classification from clinical transcripts.
- Implemented prompt-based PTSD classification using LLaMA with DSM-5-informed prompts.
- Conducted error analysis on symptom severity and depression comorbidity, revealing model sensitivity to clinical subtleties.
- Identified linguistic biomarkers and temporal patterns for depression in clinical interactions.

Automated Detection of Delusional Content in Audio Diaries Mar 2025 – now

- Designed a four-level prompt engineering framework to extract delusional themes and affective and behavioral responses from naturalistic speech transcripts.
- Implemented a multi-agent LLM pipeline (GPT-OSS, Qwen, GLM) for fine-grained multi-label annotation and benchmarked its consistency against human annotations.

LLM-based Chatbot Design for the Elderly Community Mar 2024 – now

- Designed and developed an LLM-powered chatbot prototype to improve daily life for residents of a retirement community.
- Conducted user interviews and co-design workshops with 30+ residents from the retirement community.
- Engineered prompts to ensure clear responses and implemented a voice-enabled interface to enhance accessibility and user satisfaction.

LLM-driven Development of a Digital Therapeutic for PTSD Jan 2025 – now

- Led foundational research for a PTSD mHealth app, using LLMs to synthesize evidence-based treatment components.
- Ran a human-in-the-loop workshop with experts to iteratively refine AI-generated therapeutic content.
- Used LLMs to generate technical specifications from expert-validated content for app prototyping.

Harvard Medical School | Advisors: Dr. Li Zhou & Prof. Pengyu Hong Longwood, MA

Research assistant / Project core member

Bias and Missingness Handling in Mortality Prediction for ARDS Feb 2022 – Dec 2022

- Preprocessed MIMIC-IV data and retrieved data for ARDS patients with ICU ventilation records; calculated and visualized demographic distributions and missingness rates for all features used in mortality prediction.
- Built a workflow to impute missing values based on different types of missingness using four imputation methods.
- Used propensity score matching to detect and narrow bias in the data and compared machine learning model performance for mortality prediction.
- Contributed to grant applications for the lab's work on handling bias in EHR data.

Project leader / First author | Advisors: Dr. Li Zhou & Dr. Liqin Wang

Systematic Review of Bias in Healthcare AI Using EHR Data Dec 2022 – Dec 2023

- Followed a PRISMA workflow with scripted keyword search to review and filter over 200 papers on bias handling in healthcare AI.
- Reviewed full texts and recorded evaluation metrics and detailed methods, classifying types of bias and mitigation approaches.
- First-authored a systematic review summarizing bias handling in EHR data, published in JAMIA.

Brandeis University | Advisors: Prof. Hongfu Liu & Prof. Pengyu Hong Waltham, MA

Research assistant / Project core member / Co-first author

Virus-host Interactions of SARS-CoV-2 Predicted by a Network-based Cross-species Model Mar 2020 – Aug 2020

- Researched virus-host interactions of previously published coronaviruses and compared SARS-CoV-2 with them after multiple sequence alignment in terms of domains, functions, and mutations.
- Built graphs representing sequence similarities in viral and host proteins and networks of protein-protein interaction.
- Performed graph embedding and trained a network-based cross-species deep learning model to predict high-potential PPIs and pathways of SARS-CoV-2 across animal species.

The Prediction of SARS-CoV-2 Mutation Using Deep Learning and Evolutionary Algorithms Apr 2020 – Sep 2021

- Collected data on identified SARS-CoV-2 mutations over time and location and analyzed the datasets to calculate mutation rates and identify hot zones.

- Built a model combining evolutionary algorithms and GANs to predict SARS-CoV-2 mutations from mutation history.
- Studied the influence of mutations on the 3-D structure of viral proteins.

Massachusetts Institute of Technology Cambridge, MA & Online

Research intern | Advisor: Prof. Jean-Francois Hamel

Bioreactor Simulation for Vaccine Production Optimization Jan 2020 – Aug 2020

- Learned and used bioreactor simulators to cultivate cells and study bioprocesses for both mammalian and yeast cells.
- Researched experimental settings for vaccine production, ran tests using AAV on the simulators, compared with experimental results, and optimized parameterization of COVID-19 vaccine production to maximize yield.
- Wrote weekly reports and a summary report on the optimal environment for potential expanded-scale production.

INDUSTRY & INTERNSHIP EXPERIENCES

Microsoft | Applied Scientist Intern Redmond, WA Jun 2026 – Aug 2026

Evaluating the Robustness of Enterprise AI Agents to Communication Style Variation

- Investigated the robustness of enterprise AI agents to communication-style variation, quantifying how directness, formality, verbosity, politeness, emotional tone, and grammaticality in user requests affect agent behavior.
- Developed a communication-style taxonomy and an automated generation framework that produces controlled linguistic variants of user requests while preserving semantic intent.
- Designed and executed large-scale evaluations across enterprise email, retrieval, and tool-use agent environments, measuring task completion, retrieval quality, and tool-use reliability.
- Built end-to-end experimentation pipelines covering variant generation, benchmark construction, experiment orchestration, result aggregation, and analysis.
- Quantified robustness gaps using matched-pair analysis, bootstrap confidence intervals, and significance testing, identifying communication styles that measurably degrade agent performance.
- Extended the study to multilingual and dialectal variation to assess fairness and generalization across diverse user populations, surfacing disparities that disproportionately affect non-standard language users.
- First-authored a manuscript submitted to the Workshop on Evaluation of Interactive Agents (IAEval) @ NeurIPS 2026, with an extended version in preparation for NAACL.

Amazon | Software Development Engineer Intern Bellevue, WA May 2022 – Aug 2022

- Developed and optimized shortest-path algorithms for the Supply Chain Optimization group.
- Used clustering and machine learning methods to find the optimal partition of the logistics network for transportation.
- Visualized the network before and after optimization together with the statistics the algorithm improved.

Gordian Ventures | Investment Intern Shanghai, China Mar 2023 – Aug 2023

- Researched emerging technologies including t-RNA therapeutics, aptamers, and brain-computer interfaces, and wrote a published information book.
- Mapped biotech companies funded by major ventures, summarized their key technologies and pipelines, and clustered and visualized recent trends in drug discovery.
- Estimated market size and growth rate for the peptide drug market and supported valuation of newly incubated companies using a DCF model.

SERVICE & LEADERSHIP

Local Chair, Conference on Health, Inference & Learning (CHIL) 2026 2025 – now

- Coordinating local logistics, venue, and community engagement for CHIL 2026, an annual machine learning and health conference held in Seattle.

Research Mentor, LLM-Powered Chatbot for Retirement Communities 2024 – now

- Mentored two groups of high school students in user-centered research, chatbot prototyping, data analysis, and academic writing; the collaboration produced two conference papers presented at GSA 2025 and AMIA 2026.

Peer Reviewer 2023 – now

- Reviewer for conferences and journals in NLP, machine learning, and biomedical informatics.

PUBLICATIONS & MANUSCRIPTS

Feng Chen, Ritam Dutt, Aynaz Taheri, Alex Williams, *Does It Matter How You Ask? Communication Style Variation in Retrieval and Agentic Assistants.* Accepted by the Workshop on Evaluation of Interactive Agents (IAEval) @ NeurIPS 2026; extended version submitted to NAACL.

Feng Chen, Liqin Wang, Julie Hong, Jiaqi Jiang, Li Zhou, *Unmasking Bias and Inequities: A Systematic Review of Bias Detection and Mitigation in Healthcare Artificial Intelligence Using Electronic Health Records*, published in JAMIA. DOI: 10.1093/jamia/ocae060

Feng Chen*, Hangyu Du*, Hongfu Liu, Pengyu Hong, *Network-based Virus-Host Interaction Prediction with Application to SARS-CoV-2*, published in Cell Patterns (* co-first authors). DOI: 10.1016/j.patter.2021.100242

Feng Chen, Manas Satish Bedmutha, Ray-Yuan Chung, Janice Sabin, Wanda Pratt, Brian R. Wood, Nadir Weibel, Andrea L. Hartzler, Trevor Cohen, *Toward Automated Detection of Biased Social Signals from the Content of Clinical Conversations*, oral presentation as full paper at the AMIA 2024 Annual Symposium. PMID: 40417471

Feng Chen, Dror Ben-Zeev, Gillian Sparks, Arya Kadakia, Trevor Cohen, *Detecting PTSD in Clinical Interviews: A Comparative Analysis of NLP Methods and Large Language Models*, presented at PSB 2026. DOI: 10.1142/9789819824755_0019

Feng Chen, Weizhe Xu, Changye Li, Serguei Pakhomov, Alex Cohen, Simran Bhola, Sandy Yin, Sunny X Tang, Michael Mackinley, Lena Palaniyappan, Dror Ben-Zeev, Trevor Cohen, *Reading Between the Lines: Combining Pause Dynamics and Semantic Coherence for Automated Assessment of Thought Disorder*, published in Neuropsychologia. DOI: 10.1016/j.neuropsychologia.2026.109473

Feng Chen, Justin Tauscher, Changye Li, Meliha Yetisgen, Alex Cohen, Adam Kuczynski, Angelina Pei-Tzu Tsai, Benjamin Buck, Dror Ben-Zeev, Trevor Cohen, *Automated Detection and Classification of Delusion-related Content in Naturalistic Audio Diaries Using Multi-Agent Language Models*, accepted at the ACL 2026 Workshop on Computational Linguistics and Clinical Psychology (CLPsych).

Feng Chen, Ray-yuan Chung, Yein Jeon, Oleg Zaslavsky, *Improve Technology and eHealth Literacy in a Retirement Community via the Use of LLM-Powered Chatbot*, presented at GSA 2024. DOI: 10.1093/geroni/igae098.4005

Feng Chen, Luna Xingyu Li, Ray-yuan Chung, Yein Jeon, Wenyu Zeng, Oleg Zaslavsky, *Learning from Elders: Making an LLM-powered Chatbot for Retirement Communities more Accessible through User-centered Design*, presented as full paper at CALD-AI @ ASIS&T 2025. DOI: 10.5281/zenodo.15292697

Feng Chen, Luna Xingyu Li, Ray-Yuan Chung, Wenyu Zeng, Yein Jeon, Yizhou Hu, Oleg Zaslavsky, *Bridging the Cognitive Gap: Co-Designing and Evaluating a Voice-Enabled Community Chatbot for Older Adults*, arXiv preprint: https://arxiv.org/abs/2603.11303. Accepted by the AMIA 2026 Annual Symposium as full paper.

Feng Chen, Manas Bedmutha, Janice Sabin, Andrea Hartzler, Nadir Weibel, Trevor Cohen, *Depression Detection at the Point of Care: Automated Analysis of Linguistic Signals from Routine Primary Care Encounters*, arXiv preprint: https://arxiv.org/abs/2604.06193. Accepted by the AMIA 2026 Annual Symposium as poster.

Manas Satish Bedmutha, **Feng Chen**, Andrea Hartzler, Trevor Cohen, Nadir Weibel, *Can Language Models Understand Social Behavior in Clinical Conversations?*, accepted by CHIL 2026.

Xiruo Ding, Zhecheng Sheng, Brian Hur, **Feng Chen**, Serguei VS Pakhomov, Trevor Cohen, *Enhancing Robustness of Foundation Model Representations under Provenance-related Distribution Shifts*, presented at the Workshop on Distribution Shifts, 37th Conference on Neural Information Processing Systems (NeurIPS 2023).